

John J. Guerrero

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EDUCATION

Johns Hopkins University

Ph.D. in Computer Science

Advisors: Benjamin Van Durme, Anqi Liu

Baltimore, MD

Aug 2026 – 2031 (expected)

Dartmouth College

B.A. in Computer Science, Minor in Chinese; Valedictorian, Phi Beta Kappa

Honors Thesis: *Towards Robust and Holistic Bias Benchmarking for Large Language Models* (High Honors); Advisor:

Soroush Vosoughi

Hanover, NH

Sep 2022 – Jun 2026

HONORS AND AWARDS

- **NSF Graduate Research Fellowship (2026)**: One of 260 fellows selected nationally in computer science.
- **John G. Kemeny Computing Prize (2025, 2026)**: Awarded annually for innovative computing projects by Dartmouth undergraduates; one of two two-time winners on record.
- **Dartmouth CS Department Travel Grant (2026)**: \$1,500 to present at ACL 2026.
- **Dartmouth UGR Conference Travel Grant (2026)**: Competitive \$1,000 grant to present at ACL 2026.
- **Barry Goldwater Scholarship (2025)**: One of 50 recipients nationally in computer science or math.
- **Rhodes Scholarship Finalist (2025)**: One of 238 finalists selected nationally across all fields.
- **Phi Beta Kappa (2025)**: Elected early; top 30 in the Class of 2026.
- **U.S. Congressional Citation (2025)**: Awarded by a member of US Congress for my research accomplishments.
- **Meta NeurIPS Travel Award (2024)**: Awarded by Meta to attend NeurIPS 2024 for work on model calibration.
- **Francis L. Town Scientific Prize (2024)**: Awarded at the end of sophomore year to one student in each scientific department at Dartmouth.
- **James O. Freedman Presidential Scholar (2024)**: Research assistantship reserved for top Dartmouth students preparing for a senior honors thesis.
- **Rufus Choate Scholar (2023 – 2026)**: Awarded to the top 5% of Dartmouth undergraduates for academic excellence; received all four years.
- **U.S. Presidential Scholars Program Semifinalist (2022)**: One of 620 nationally selected for academic excellence, community service, and leadership.
- **Sharon K. Harvey Memorial Scholarship (2022)**: One of two recipients; for exemplary academic performance, leadership, and community service.
- **JHU APL ASPIRE Scholarship (2022)**: One of two interns program-wide, for research contributions.

PUBLICATIONS

* *denotes equal contribution*

- [1] **When Debiasing Backfires: Counterintuitive Side Effects of Preprocessing-Based Stereotype Mitigation**
Yahan Zheng, **John J. Guerrero**, Soroush Vosoughi, Weicheng Ma
Findings of ACL 2026 [paper]
- [2] **Scalable and Culturally Specific Stereotype Dataset Construction via Human-LLM Collaboration**
Weicheng Ma*, **John J. Guerrero***, Soroush Vosoughi
EMNLP 2025 [paper]
- [3] **Decision-Driven Calibration for Cost-Sensitive Uncertainty Quantification**
Gregory Canal, Vladimir Leung, **John J. Guerrero**, Philip Sage, I-Jeng Wang
NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty [paper]
- [4] **Labeling Software Security Vulnerabilities**
Irena Bojanova, **John J. Guerrero**
IEEE IT Professional, 2023 [paper]

[5] **LitCovid in 2022: An Information Resource for the COVID-19 Literature**

Qingyu Chen*, Alexis Allot*, Robert Leaman, Chih-Hsuan Wei, Elaheh Aghaari, **John J. Guerrerio**, Lilly Xu, Zhiyong Lu
Nucleic Acids Research, 2023 [paper]

RESEARCH EXPERIENCE

Transparent, Reliable, Illustrative & Practical AI Lab, Dartmouth College

Hanover, NH

Machine Learning Research Assistant

Jun 2025 – Jun 2026

Mentors: Yaoqing Yang

- Developed Bayes Pruning Prior (BPP), a Bayesian weight-editing method to enhance LLM in-context reasoning by reducing reliance on generic, memorized associations that override information in the prompt. Outperformed all baselines, including LASER and fine-tuning, on Pythia LLMs.
- Developed a parameter-efficient variant of Born-Again Networks, matching full Born-Again training on ResNet, BERT, and EfficientNet.

Minds, Machines, and Society Group, Dartmouth College

Hanover, NH

Machine Learning Research Assistant

Mar 2024 – Jun 2026

Mentors: Soroush Vosoughi, Weicheng Ma

- Created MOSAIC, a bias benchmark that measures the semantic content of LLM generations rather than preferences between fixed sequences. Introduced four new bias-measurement metrics and theoretically principled measures of benchmark robustness and data efficiency.
- Showed that common data-level debiasing (e.g., removing or swapping group mentions in training data) can increase stereotyping toward other, non-targeted groups, side effects that standard benchmarks miss [1].
- Designed a human–LLM collaborative annotation framework to reduce the cost of developing bias benchmarking datasets for low-resource languages. Created EspanStereo (2,690 instances of 538 stereotypes), the first native Spanish bias benchmark instantiated in natural language [2].

Security and Privacy in the Lifecycle of IoT Group, Dartmouth College

Hanover, NH

James O. Freedman Presidential Scholar

Sep 2024 – Nov 2025

Mentors: Tim Pierson, O. Sami Saydjari

- Designed a device-agnostic privacy-preference encoding that lets smart-home visitors express their preferences. Built a benchmark evaluating how well frontier LLMs translate users’ natural-language privacy preferences into this encoding, a key step toward agentic systems that enforce those preferences automatically.

Analytic Capabilities Group, Johns Hopkins Applied Physics Laboratory

Laurel, MD

Machine Learning & AI Intern

Jun 2025 – Aug 2025

Mentors: I-Jeng Wang, Greg Canal

- Developed theoretically grounded methods to estimate model competency, the probability that a given prediction is correct.
- Built monitoring infrastructure to ensure the reliability of a new platform for mission-critical AI.

Machine Learning & AI Intern, extended into the academic year

Jun 2024 – Jan 2025

Mentors: I-Jeng Wang, Greg Canal

- Designed a calibration method that tunes a model’s confidence to minimize the cost of a decision-maker’s downstream choices; validated in a simulated search-and-rescue scenario [3], then extended to recidivism prediction to mitigate harm from biased decision-makers whose policies are unknown.

Software Quality Group, National Institute of Standards and Technology

Gaithersburg, MD

Summer Undergraduate Research Fellow

Jun 2023 – Dec 2023

Mentors: Irena Bojanova

- Formally modeled 200+ software weaknesses and 10+ vulnerabilities, creating a structured resource for AI-driven vulnerability analysis.
- Identified 11 ambiguities in the current vulnerability classification standard that cause misclassifications, and proposed fixes [4].

Biomedical Text Mining Group, National Institutes of Health

Bethesda, MD

Summer Intern

Jun 2022 – Aug 2022

Mentors: Zhiyong Lu, Qingyu Chen

- Built BERT-based named-entity recognition models to identify vaccine and strain names across 300,000+ papers in LitCovid; reached $F1 = 0.97$ (+0.06 over prior models) and was deployed in LitCovid [5].

SERVICE

- **Program Committee (2026):** Workshop on Stereotypes Across Cultures in Language Technologies at ACL 2026
- **Secondary Reviewer (2024 – 2026):** ACL Rolling Review

OTHER EXPERIENCE

Digital Applied Learning & Innovation (DALI) Lab, Dartmouth College Hanover, NH
Data Mentor, Machine Learning Mar 2025 – May 2026

Partners: Yaoqing Yang (ML Diagnostics); Dartmouth-Hitchcock Medical Center (Smart Microscope)

- *ML Diagnostics*: Led a team of 5 to predict a model's double-descent phase from the loss landscape of a single checkpoint; reached 91% accuracy transferring from CIFAR-10 to CIFAR-100 and, when guiding hyperparameter tuning, was competitive with Bayesian optimization using half as many tested configurations.
- *Smart Microscope*: Led a team of 3 engineers to build a virtual staining model that turns images of unstained pathology slides into H&E-style stained images; 14% more accurate than prior CNN baselines.

Data Developer, Machine Learning Sep 2024 – Mar 2025

Partner: National Park Service

- *Tide Pool*: Built and deployed image segmentation models to automate analysis of tide pool samples.

Cornerstone Research Boston, MA
Winter Analyst Jan 2025 – Mar 2025

- Analyzed the AI industry to forecast the market and societal impacts of a potential antitrust action, informing expert testimony.

OUTREACH AND LEADERSHIP

Dartmouth CoderDojo Hanover, NH
Club Executive & Visit Lead; previously Volunteer Teacher Sep 2023 – Nov 2025

- Designed the club's first Scratch curriculum and led visits teaching programming at under-resourced local schools; co-wrote a successful \$1,000 grant to expand robotics lessons.

Norwalk Housing Authority Student Success Program Norwalk, CT
Volunteer Math Tutor Jun 2023 – Jun 2024

Dartmouth Center for Social Impact, SIBS Program Hanover, NH
Mentor Mar 2023 – Jun 2024